



Machine Learning–Based Forecasting of Chlorophyll-a in Lake Munson Under Extreme Drawdown Conditions Using Gradient Boosting Regression

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Introduction

Problem Statement

Harmful algal blooms pose a significant threat to freshwater ecosystems, driven by excess nutrients like nitrogen and phosphorus. These blooms can deplete oxygen, harm aquatic life, and produce toxins, jeopardizing both environmental and human health, which is why we decided to do a study on developing an advanced predictive model for chlorophyll-a concentrations in Lake Munson to enable early warning systems and inform proactive management strategies for preventing harmful algal bloom events.



Figure 1. Lake Munson post drawdown [1]



Figure 2. Harmful algal bloom abatement in Lake Munson. [2]

Lake Munson (History)



Lake Munson, a eutrophic lake in Tallahassee, Florida, experienced severe algal blooms, leading to a complete drawdown—a process where the water level was significantly lowered. **Figure 3.** State of Lake Munson during the Drawdown [3] This intervention was necessary not only to improve the lake's condition but also to protect interconnected waterways like Ames Sink and Wakulla Springs.

Water Quality Data

- Water quality data for Lake Munson was sourced from the Leon County spanning from 2020 to 2025.
- This dataset includes measurements for total nitrogen, total phosphorus, and chlorophyll-a concentrations.
- Data collection occurred at specific, fixed sampling stations across the lake (Figure 4), rather than as a continuous time series or a full spatiotemporal grid.
- “With the extension of the drawdown at the March 21, 2023 meeting, the Board approved the addition of a Drawdown Water Quality Study to the Action Plan. This study supplemented the County’s quarterly water chemistry sampling to attain monthly water quality data in Munson Slough both upstream and downstream of the lake” (Leon County Action Plan 2023). Data sampling swapped from bimonthly to monthly due to this.

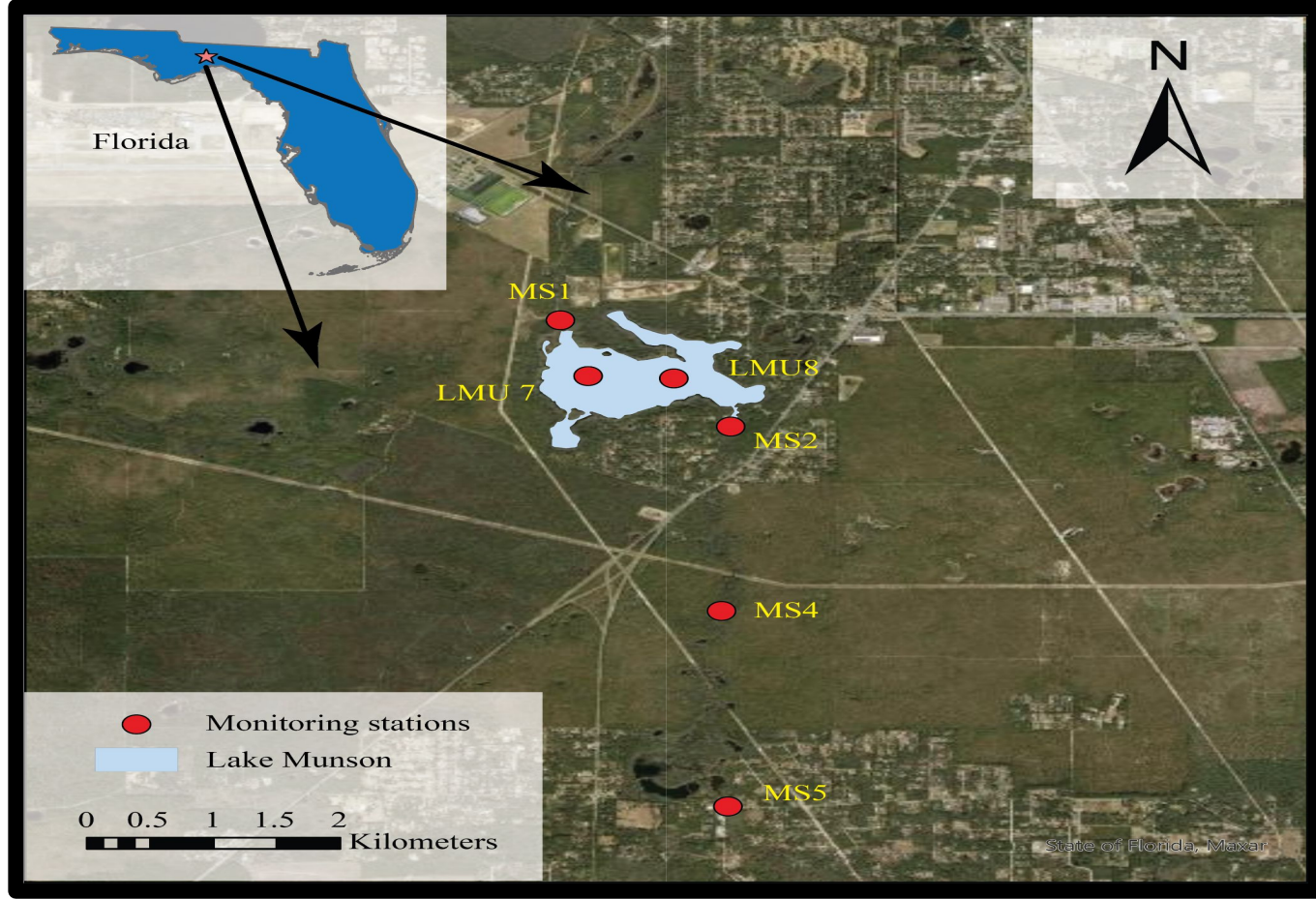


Figure 4. Sampling stations at Lake Munson. Data was collected at discrete points (MS1, MS2, MS4, MS5, LMU7, LMU8) by Wei Mao]

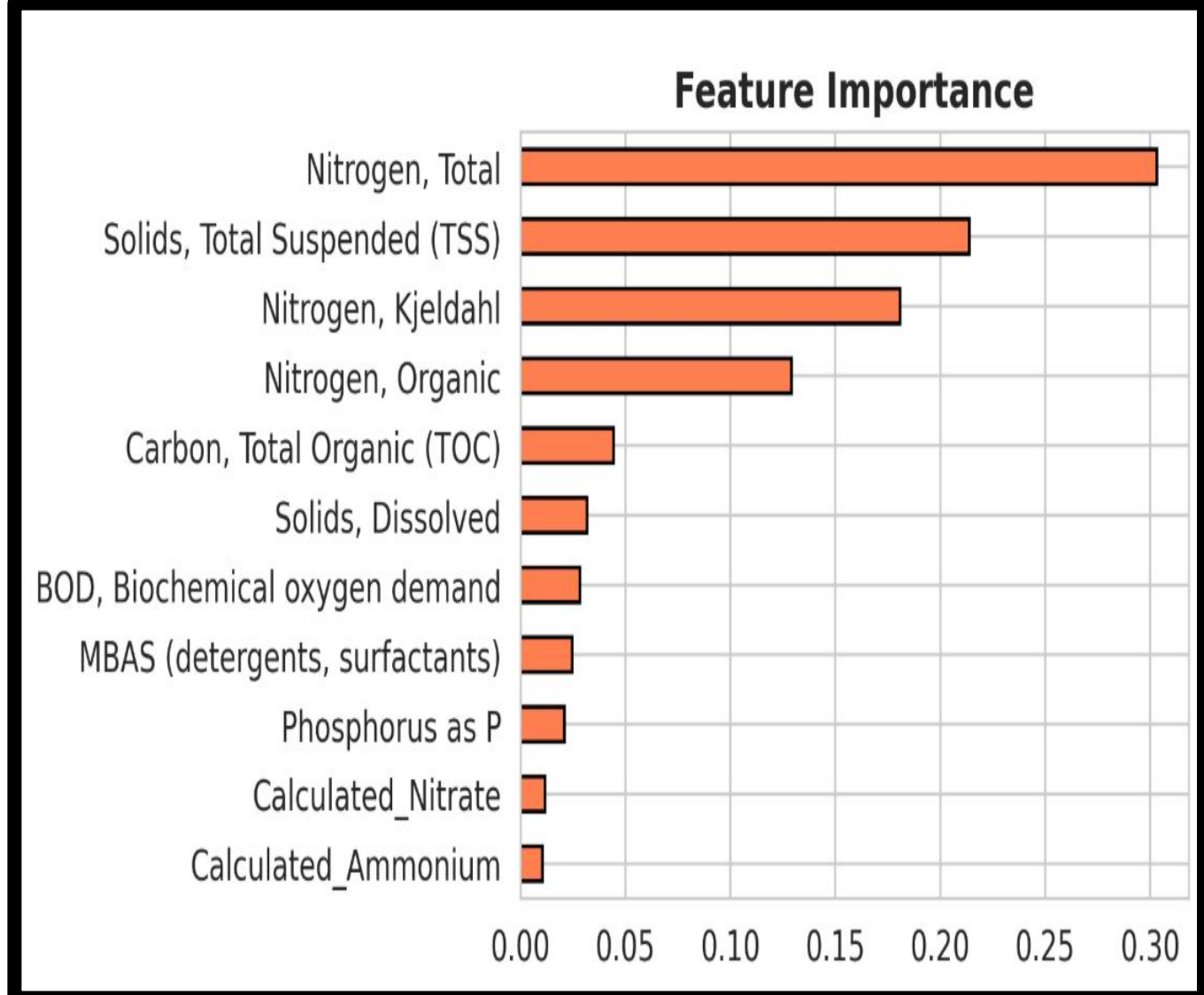


Figure 5. Feature Importance test on data[Figure created by Alex Wang]

Methodology and Data

Gradient Boosting Regressor Method

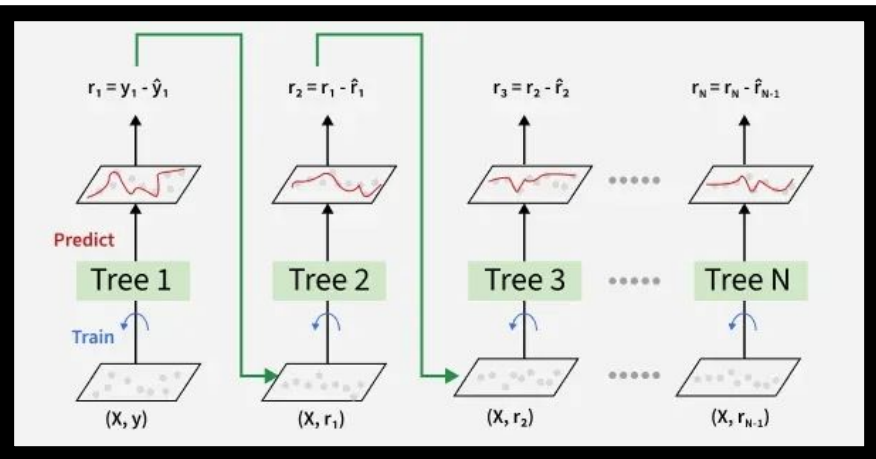


Figure 6. Gradient Boosting Regressor visualized [4]

$$F_0(x) = \underset{\gamma}{\operatorname{argmin}} \sum_{i=1}^n L(y_i, \gamma)$$

Figure 7. Formula of first prediction made by model

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x) = F_{m-1}(x)}$$

Figure 8. Formula for calculating residual in observed vs predicted

$$F_m(x) = F_{m-1}(x) + \eta \cdot h_m(x)$$

Figure 9. Scaling the next prediction based on residual

The model predicts Chlorophyll-a levels by building an ensemble of 200 sequential decision trees. Unlike models that train in parallel Gradient Boosting functions like a relay race where each new tree is specifically trained to correct the residual errors (mistakes) made by the previous trees

Baseline Initialization: The model starts with a constant guess (F0), which is the mean of all Chlorophyll-a observations:

Iterative Error Correction: For each tree (m), the model calculates the pseudo-residuals (r_{im}), representing the difference between actual and predicted values: r_{im} = −[∂F(x_i)]

Additive Update: A new tree (h_m) is fitted to these residuals and added to the model, scaled by a learning rate (η=0.1) to prevent overfitting:

Feature Importance

$$\text{Importance}(j) = \frac{1}{M} \sum_{m=1}^M \sum_{i \in T_m} \text{Improvement}_i(x_j)$$

Figure 10. Equation for computing the importance of feature j

The model utilizes the Mean Decrease in Impurity (MDI), which measures the total reduction in Mean Squared Error (MSE) provided by a specific feature.

Data Cross Fold Validation

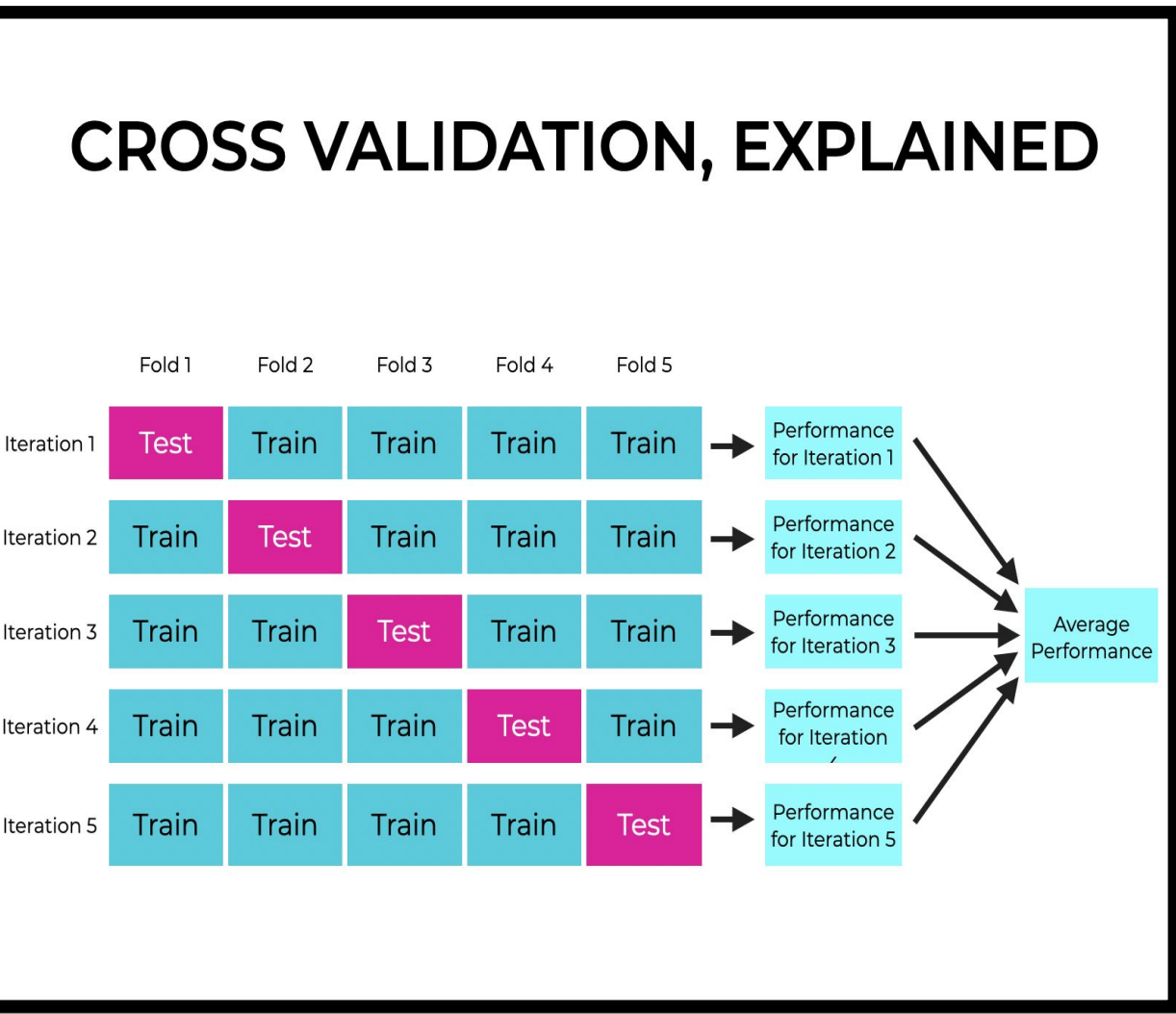


Figure 11. 5 Fold Cross Validation visualized [5]

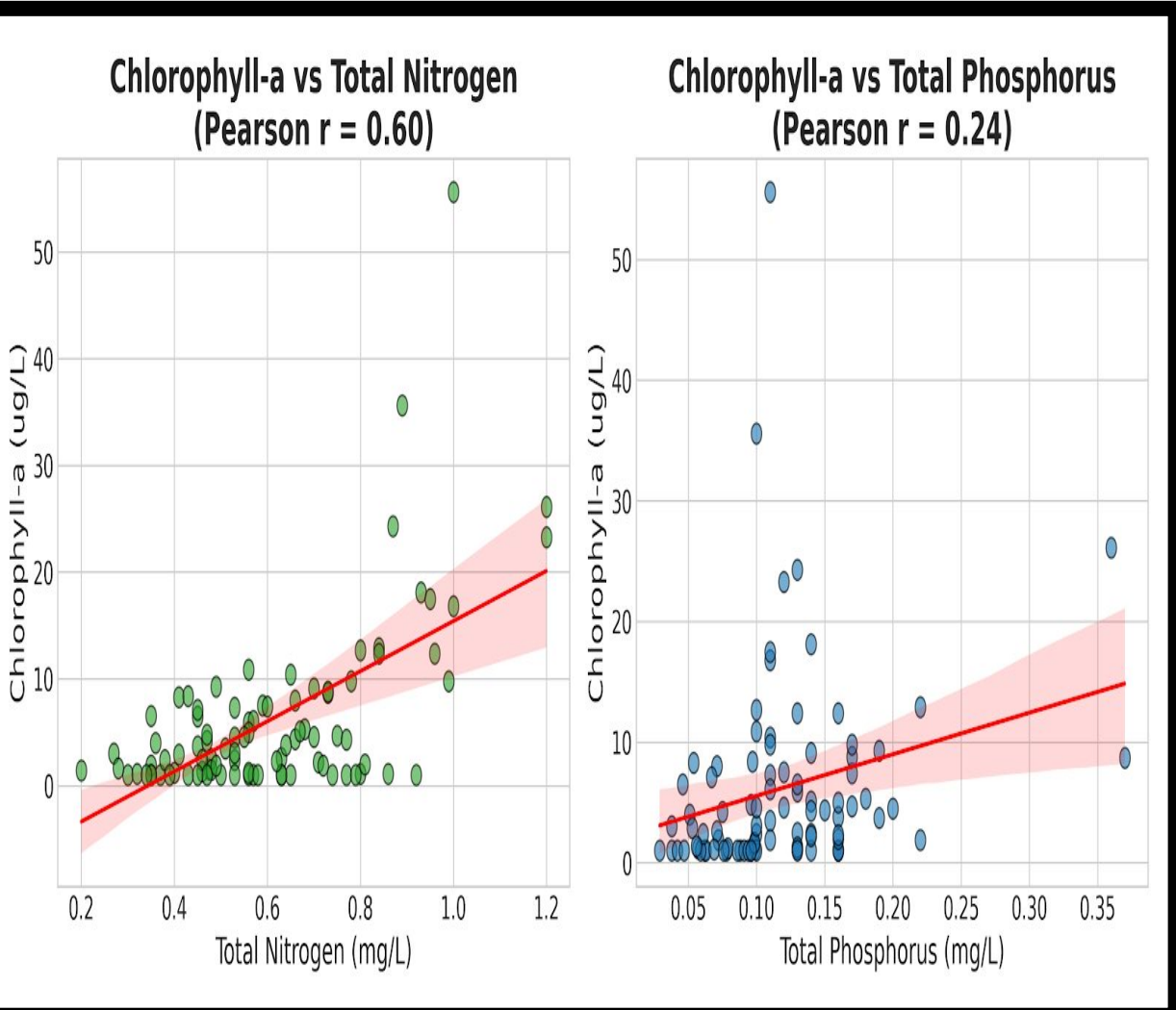


Figure 12. Feature correlation

Because the Munson Slough data is non-Gaussian and skewed (with extreme algal bloom outliers), a single random split might accidentally put all extreme values in one set. CV prevents this by ensuring the model handles the full range.

This scatter plot demonstrates a moderate positive correlation between total nitrogen + phosphorus concentrations and chlorophyll a levels.

Results

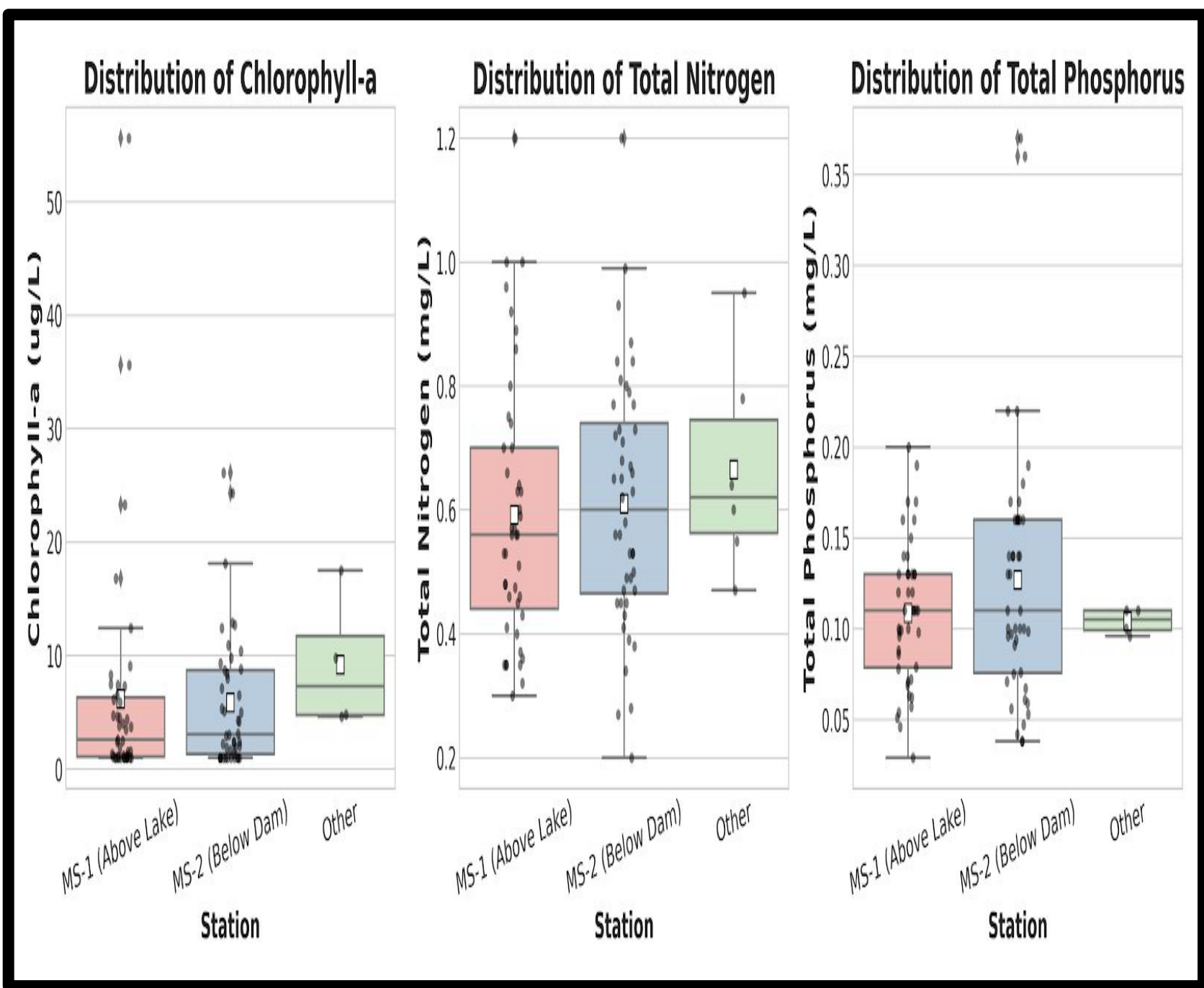


Figure 13. Summary of water quality data

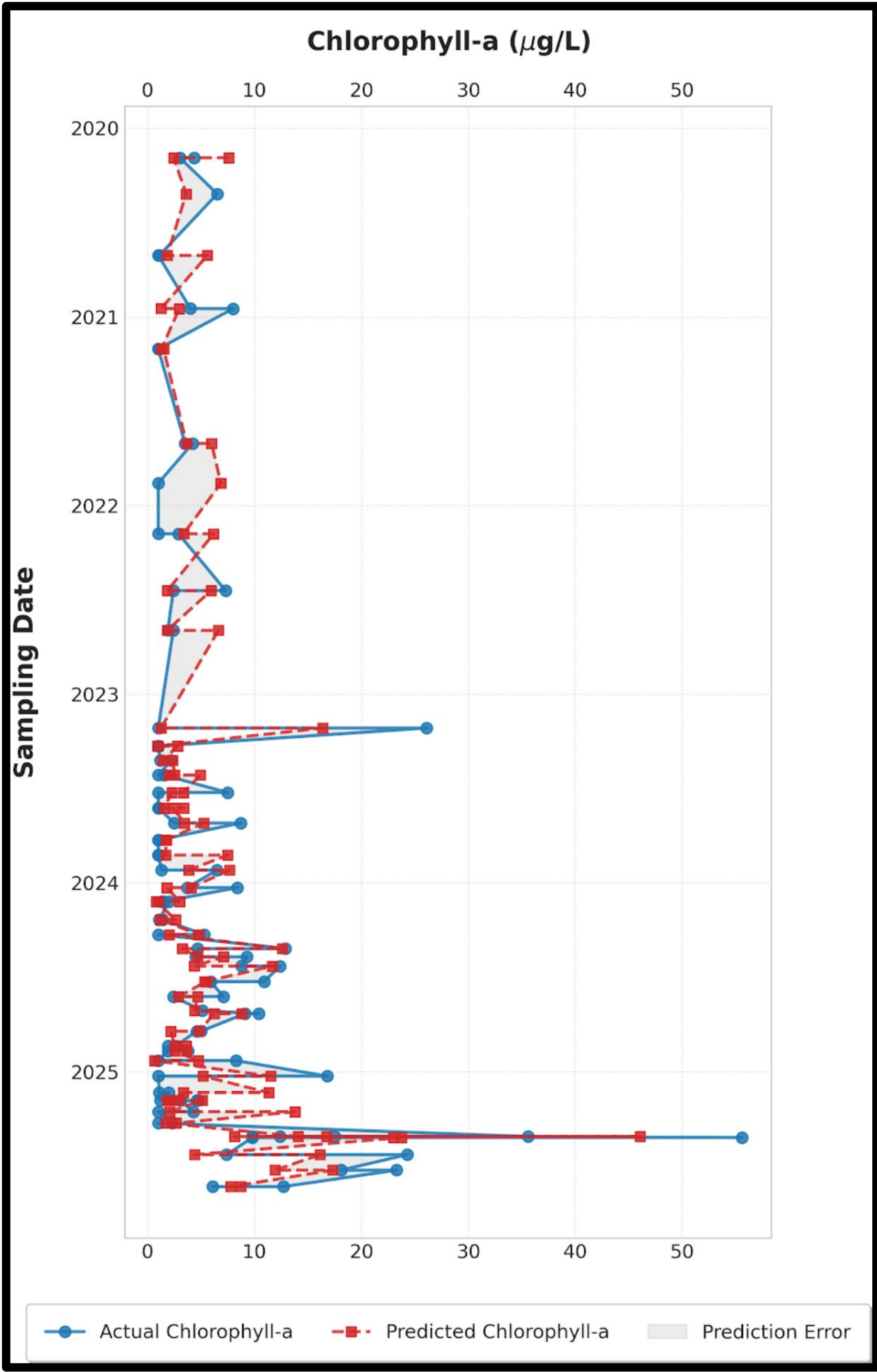


Figure 14. Model Simulation vs Actual data of Chlorophyll-a

Conclusions and Discussion

- The Lake Munson dataset indicates that model residuals follow a non-Gaussian, skewed distribution, likely due to the highly variable nature of algal blooms.
- The Gradient Boosting Regressor was utilized to address these complexities, achieving the following results: Cross-Validated Performance (5-Fold CV): R² Score: 0.4111 RMSE: 6.1864 µg/L While sampling intervals reflect significant data gaps across the stations, the Gradient Boosting model demonstrates resilience in handling the irregular temporal variations.
- Current performance suggests that future iterations should prioritize specialized small-data management practices, such as log transformation of target variables, to better account for skewness.

References

[1] <https://www.lakemunson.org/current-conditions>
[2] <https://wakullaspringsalliance.org/wp-content/uploads/2020/06/Intact-Cellular-Harvesting-for-HAB-Abatement-in-Lake-Munson.pdf>
[3] <https://www.tallahassee.com/story/news/2022/12/05/as-county-draws-down-lake-munson-residents-and-scientists-want-more/10726864002/>
[4] <https://www.geeksforgeeks.org/machine-learning/ml-gradient-boosting>
[5] <https://python.plainenglish.io/cross-validation-in-machine-learning-enhancing-model-performance-with-confidence-39402f6ca60f?gi=9e5fb211df87>